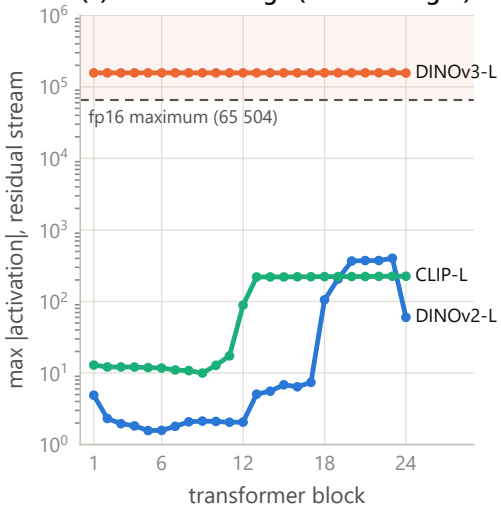
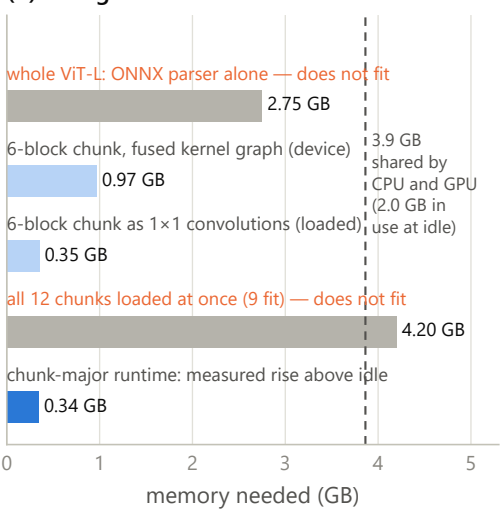

(a) activation range (64 test images)
max |activation|, residual stream
transformer block
DINOv3-L
fp16 maximum (65 504)
CLIP-L
DINOv2-L
10^6
10^5
10^4
10^3
10^2
10^1
10^0
1
6
12
18
24


(b) fitting a ViT-L into the Nano
whole ViT-L: ONNX parser alone — does not fit
2.75 GB
6-block chunk, fused kernel graph (device)
0.97 GB
6-block chunk as 1×1 convolutions (loaded)
0.35 GB
3.9 GB shared by CPU and GPU (2.0 GB in use at idle)
all 12 chunks loaded at once (9 fit) — does not fit
4.20 GB
chunk-major runtime: measured rise above idle
0.34 GB
0
1
2
3
4
5
memory needed (GB)